

WEEK 5-6

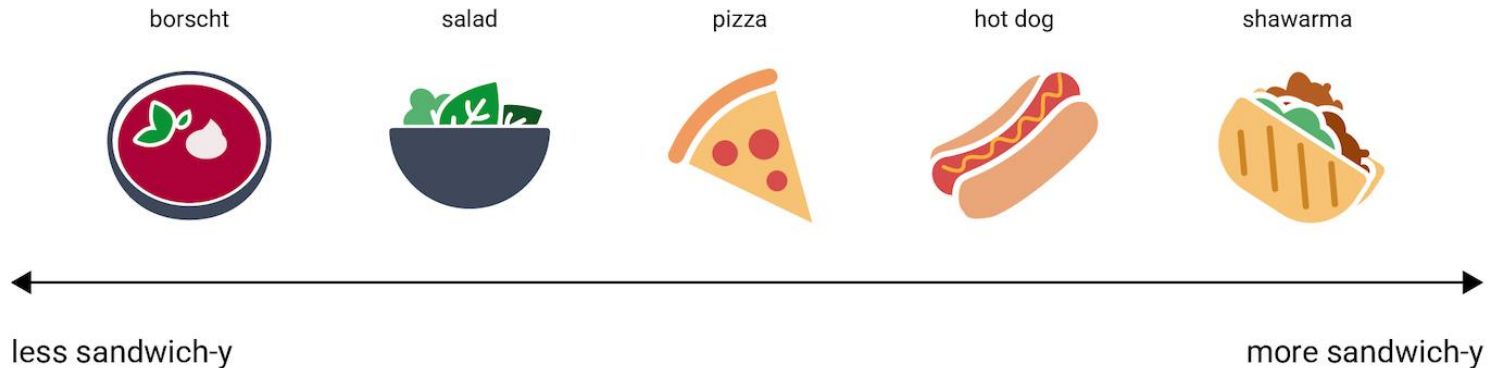
Dimension Reduction



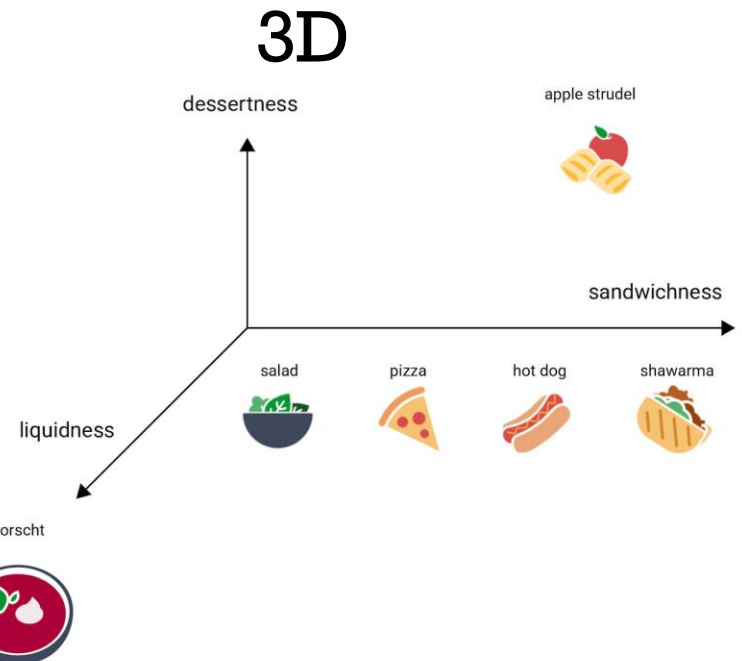
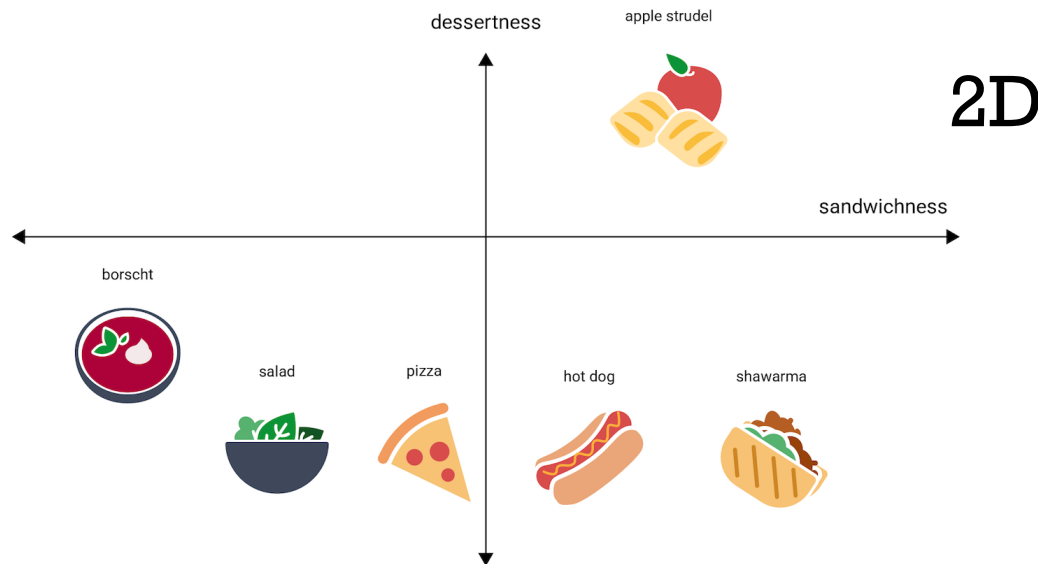
IDEA: EMBEDDING

Food Image

1D
representation



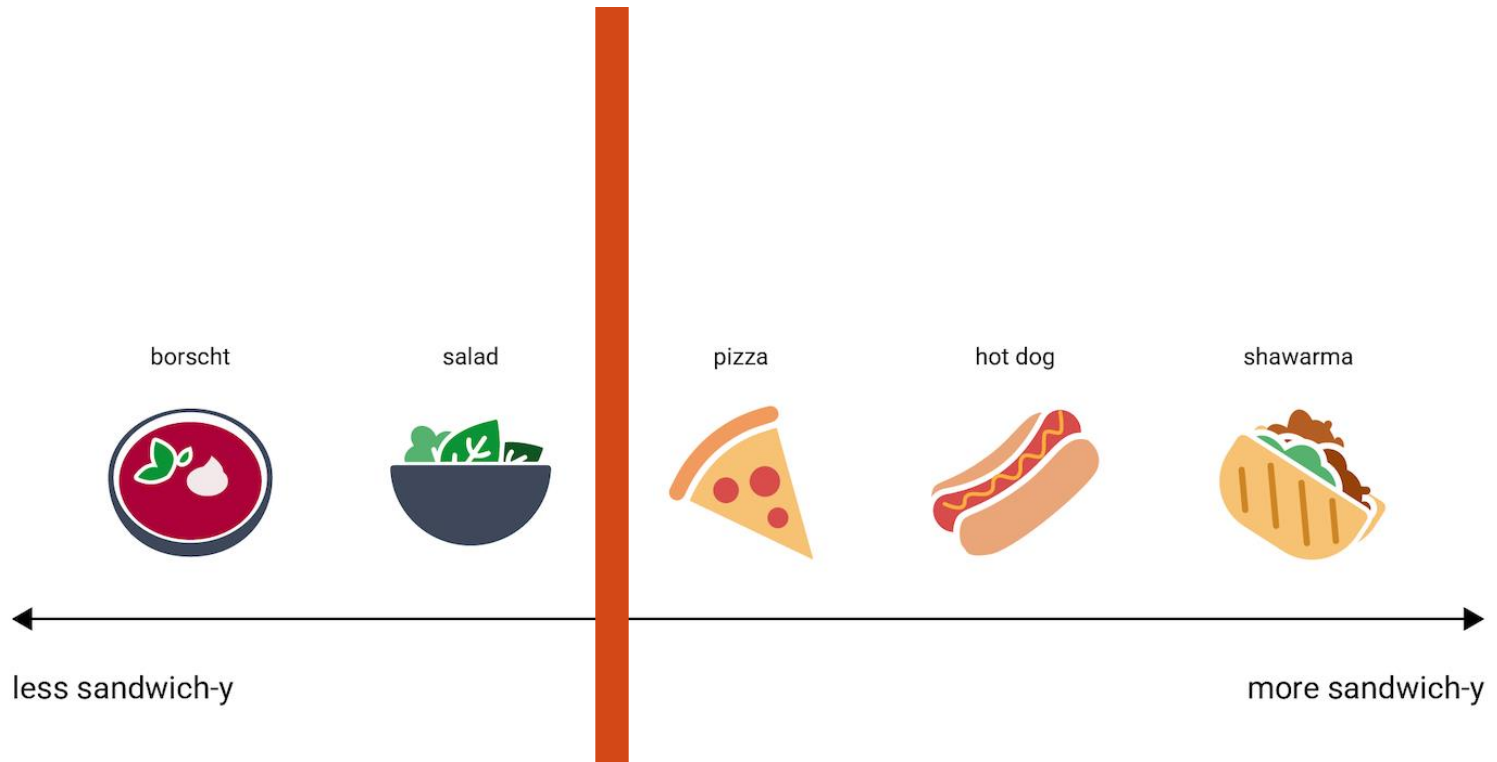
MORE DIMENSION REPRESENTATION



High dimensional data creates curse of dimensionality



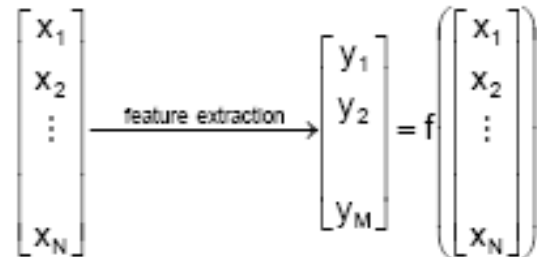
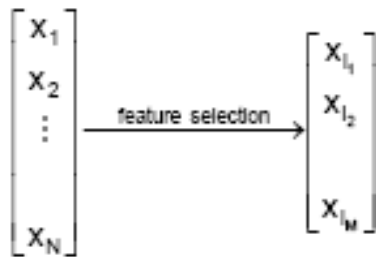
GOAL TO MAXIMIZE THE SEPARATION



FEATURE EXTRACTION

How to create the low dimension from high dimension space

1. Select some feature from your space
2. Create a transform function for your features



FEATURE EXTRACTION

- Translate/Transform into a lower dimensional space
- The transformation function can be linear transformation

$$y = Wx$$

- Or non linear, which is more common in real data



FISHER DISCRIMINATION ANALYSIS

- Maximised $J(w) = \frac{w^T S_B w}{w^T S_W w}$

Where

- $S_B = c (\mu_c - \mu)(\mu_c - \mu)^T$ is a between class scatter matrix
- $S_W = \sum_{i \in c} (x_i - \mu_c)(x_i - \mu_c)^T$ is a within class scatter matrix
- Maximising w to get the w and transforming data to one dimension
- Assumes Gaussian distribution of the data
- Solution for binary class $w^* = S_W^{-1}(\mu_1 - \mu_2)$



QUICK PROVE

Use a Lagrange multiplier λ for the denominator constraint:
(to avoid it become too large)

$$\mathcal{L}(w, \lambda) = [w^T(\mu_1 - \mu_2)]^2 - \lambda(w^T S_W w - 1)$$

- Take gradient and set to zero:

$$\nabla_w \mathcal{L} = 2(w^T(\mu_1 - \mu_2))(\mu_1 - \mu_2) - 2\lambda S_W w = 0$$

- Rearranged:

$$\begin{aligned} S_W w &\propto (\mu_1 - \mu_2) \\ \Rightarrow w^* &\propto S_W^{-1}(\mu_1 - \mu_2) \end{aligned}$$



COMPARISON

- Bayes Theorems

$$P(C_k | x) = \frac{p(x | C_k)P(C_k)}{p(x)}$$

where

$$p(x | C_k) = N(x | \mu_k, \Sigma)$$

- Discriminant Function

$$\ln \frac{P(C_1 | x)}{P(C_2 | x)} = w^T x + w_0$$

$$w^* = S_W^{-1}(\mu_1 - \mu_2)$$

- Logistic Regression

$$P(C_1 | x) = \frac{1}{1 + \exp(-w^T x - w_0)}$$

Find w that minimise the cross-entropy



DEMO

- https://github.com/AjMing/Pattern-EGCI463/blob/main/week6/Fisher_LDA.ipynb



PRINCIPAL COMPONENT ANALYSIS

- Unsupervised approach
- No class information is required
- The main idea is to reduce dimension using it's pattern



PRINCIPAL COMPONENT ANALYSIS

- Linear projection technique

$$y = \sum_{i=1}^m w_i x_i + \sum_{i=m+1}^d w_i x_i$$

- The quality of projection is measured by

$$E = \frac{1}{2} (x - y)^2 = \frac{1}{2} \sum_{i=M+1}^d u^T \Sigma u = \frac{1}{2} \sum_{i=M+1}^d \lambda$$



PRINCIPAL COMPONENT ANALYSIS(PCA)

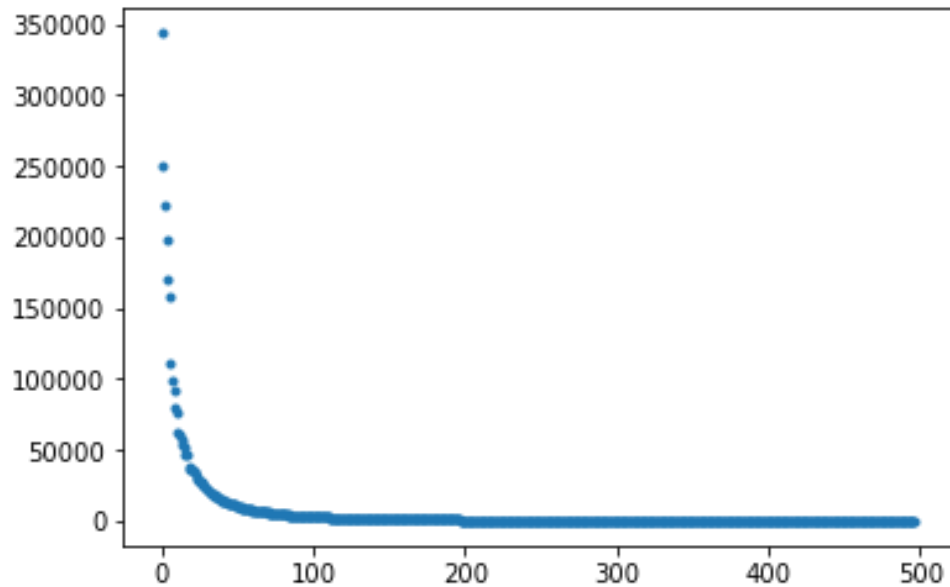
- Linearly squeeze the dimension upon to Eigen vectors axes.

$$\Sigma v = \lambda v \Rightarrow |\Sigma - \lambda I| = 0$$

- If data is non-linearly separated, it's likely to deform the data(no guarantee of good reparability)
- No class info. is attached.

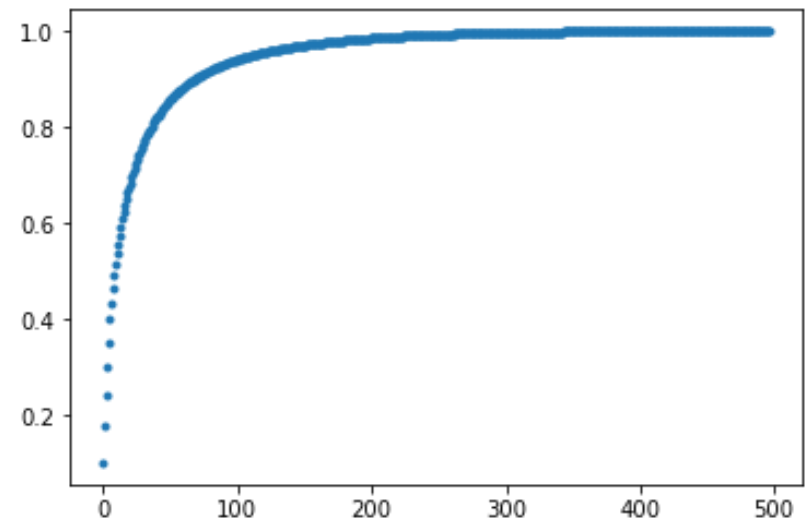


EXAMPLE : EIGEN VALUES



Eigen values

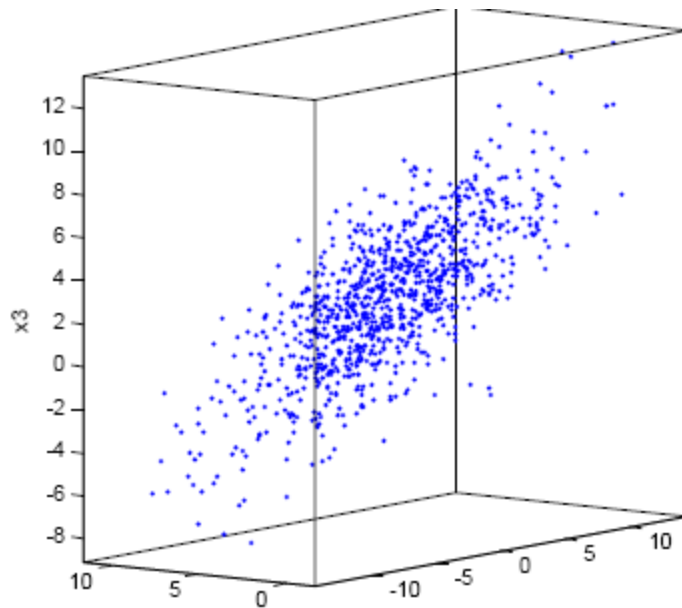
v.s.



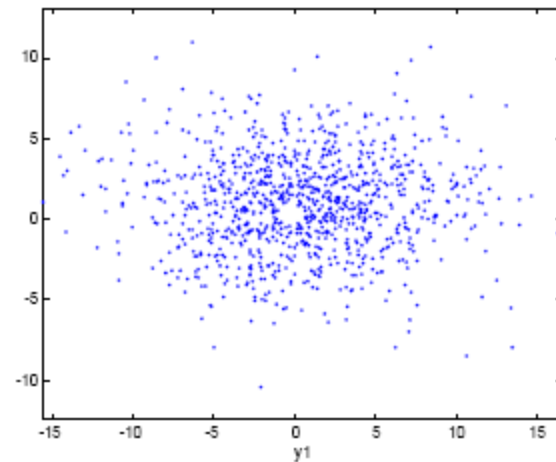
Cumulative sum eigen values



EXAMPLE



**Largest Eigen value
Projection in 2D**

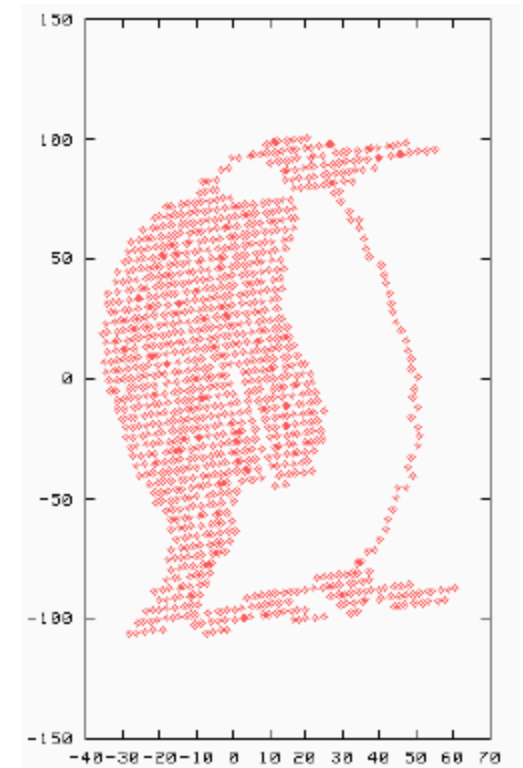
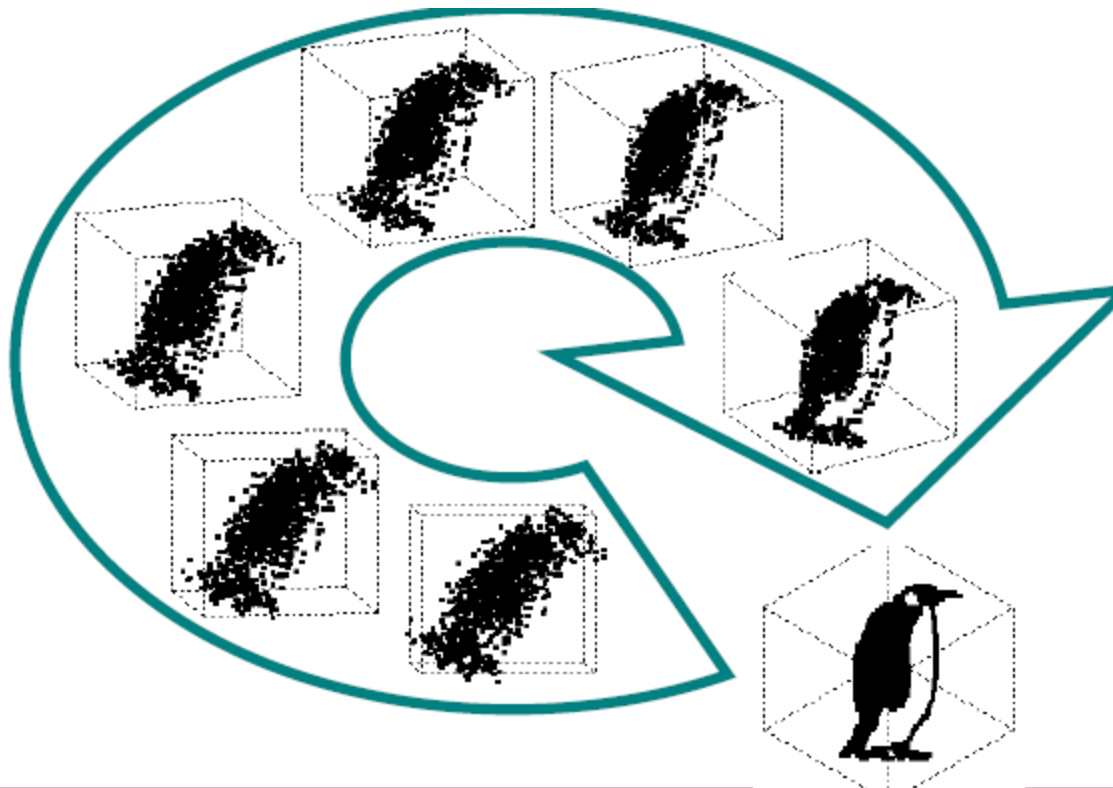


Dimensional data

3



THE USE OF PCA



*** From

Introduction to Pattern Analysis
Ricardo Gutierrez-Osuna
Texas A&M University



WHAT'S NEXT WITH PCA

- The selected space can be used with the input of
 - Neural Network
 - K-NN
 - Histogram
 - Bayesian Approaches



PCA CALCULATION

- 1. Calculate the covariance across all the data
- 2. Find the eigen values and list them descending or identify number of M manually

$$y = \sum_{i=1}^M (x_i - \mu) * v_i$$

- 3. Use the new projected data for pattern recognition
- 4. Reconstructed images can be done by

$$\tilde{x} = (y * v_{1:M}) + \mu$$



PCA CODING

```
from sklearn.decomposition import PCA
```

1. Find the first N eigen values or with percentage

```
pca = PCA(n_components=com) or pca=PCA(.percent)
```

2. Project the data with

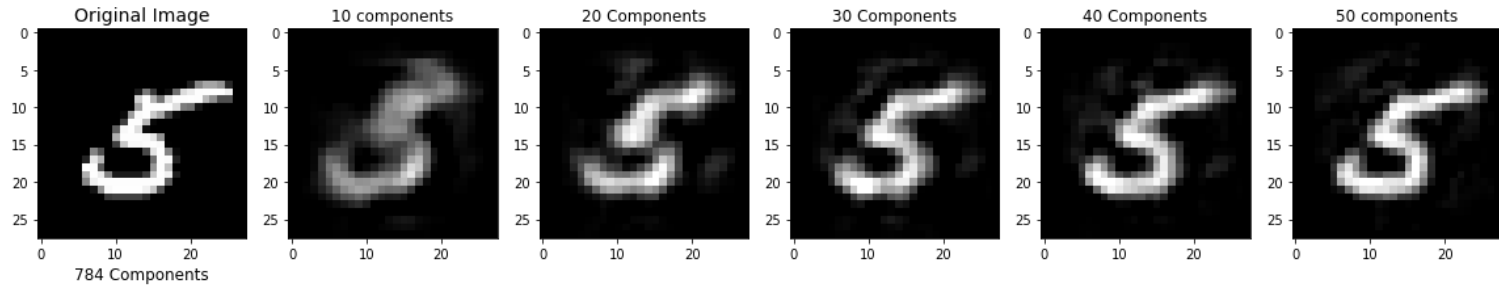
```
components = pca.transform(images)
```

3. Reconstruct the data by

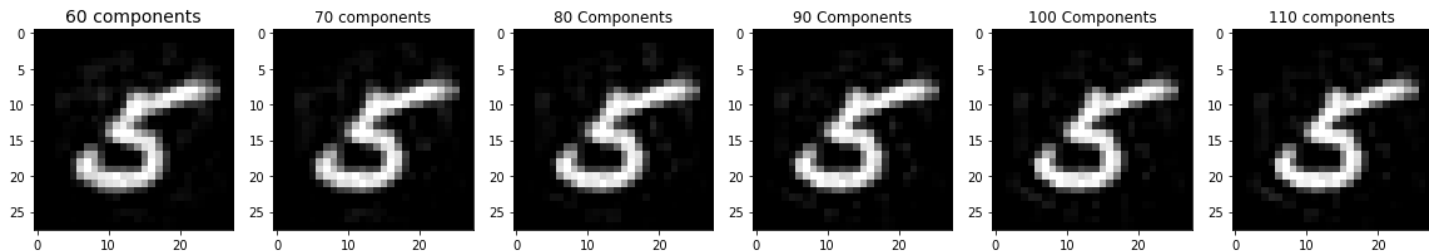
```
Original = pca.inverse_transform(components)
```



PCA RECONSTRUCTION



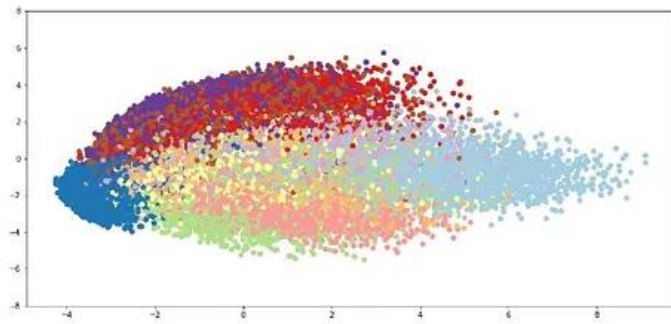
Hand written digit



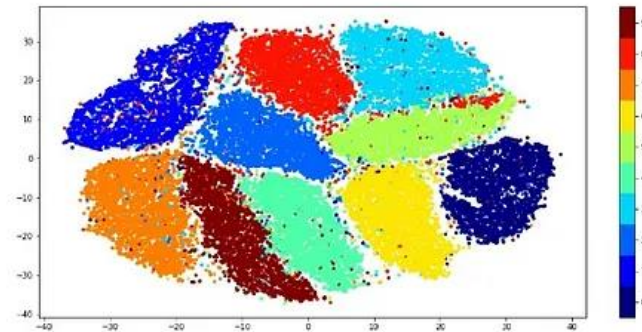
NON LINEAR EMBEDDING

t-distributed
stochastic neighbor embedding

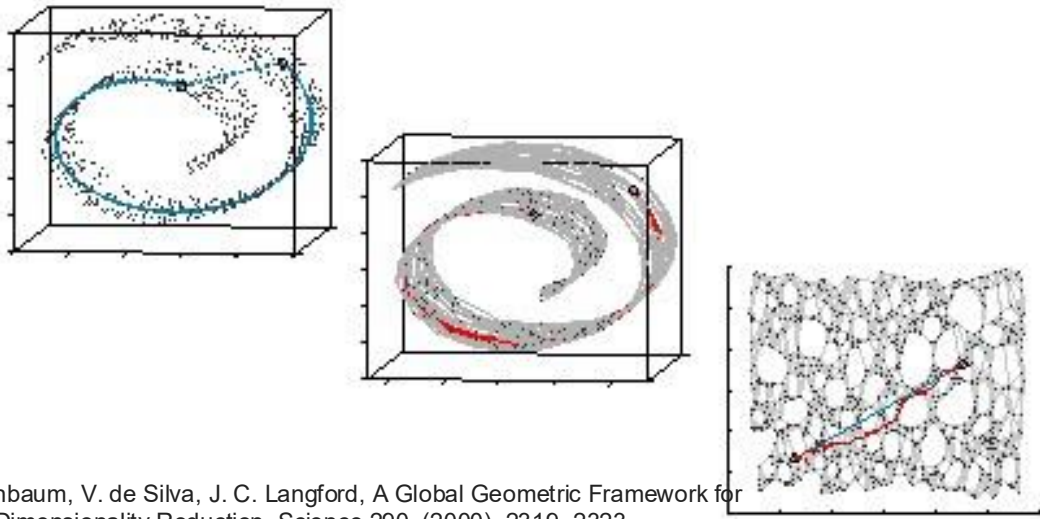
MNIST - PCA



MNIST - TSNE

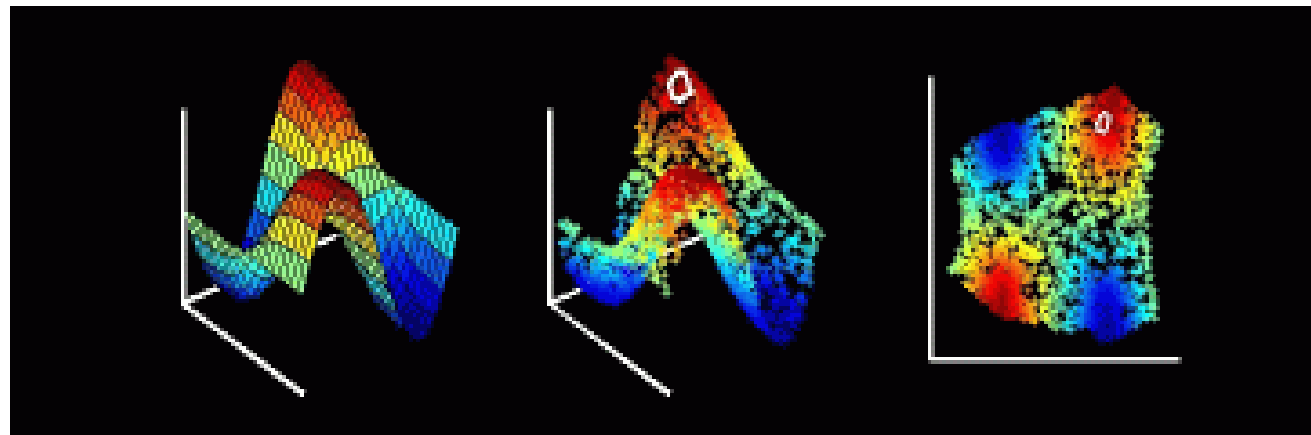


ISOMAP



J. B. Tenenbaum, V. de Silva, J. C. Langford, A Global Geometric Framework for Nonlinear Dimensionality Reduction, Science 290, (2000), 2319–2323.

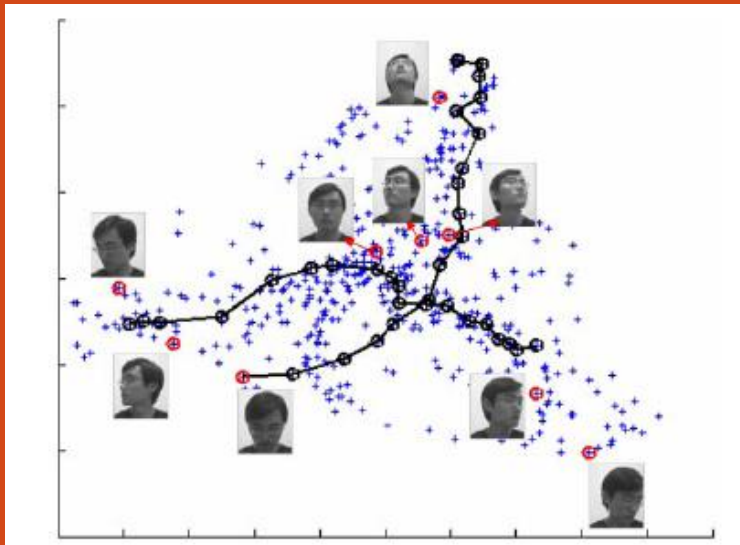
LLE



S. T. Roweis and L. K. Saul, Nonlinear Dimensionality Reduction by Locally Linear Embedding, Science Vol 290, 22 December 2000, 2323–2326.

EXAMPLES

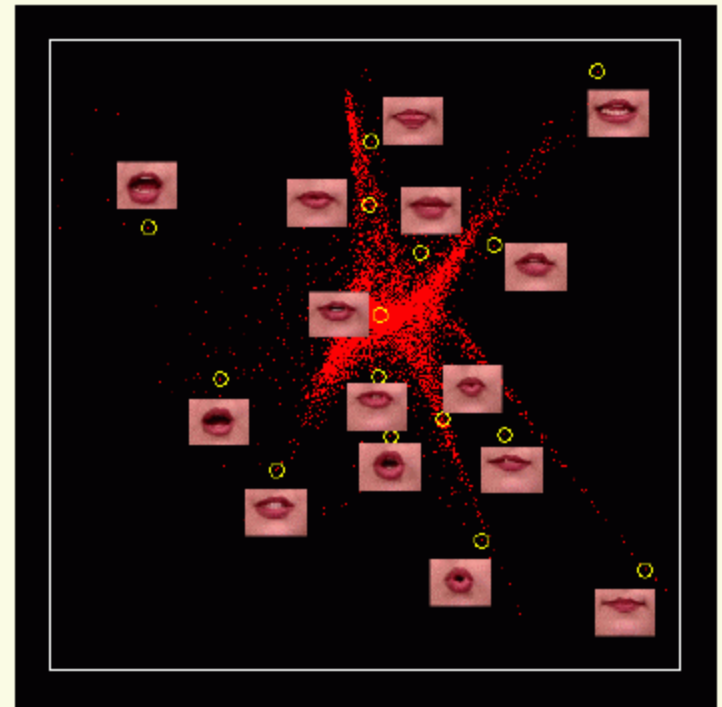
ISOMAP to visualise face angles



Xin Geng, De-Chuan Zhan, and Zhi-Hua Zhou

2010

LLE for mouth position



S.Roweis and L.K. Saul, 2000

EXAMPLE

- <https://projector.tensorflow.org/>

